FemoraHealth: A Machine Learning Framework for Polycystic Ovary Syndrome (PCOS) Prediction Using Clinical, Hormonal, and Lifestyle Features

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Abstract—Polycystic Ovary Syndrome (PCOS) is a prevalent endocrine disorder affecting women of reproductive age worldwide. Early and accurate diagnosis is critical for effective clinical management. This paper presents FemoraHealth, a comprehensive machine learning-based framework for PCOS binary classification using a dataset of 541 patients with 42 clinical, hormonal, and lifestyle features. Six classifiers—Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and XGBoost—were evaluated following rigorous data preprocessing, exploratory data analysis (EDA), multi-method feature selection, and SMOTE-based class balancing. The Random Forest model achieved best-in-class performance with accuracy 91.7%, F1-score 0.87, and AUC 0.96. SHAP (SHapley Additive exPlanations) analysis identified Total_Follicles, Symptom_Score, and Skin_Darkening as the most diagnostically significant predictors. The proposed framework demonstrates the potential of interpretable machine learning in supporting clinical decision-making for PCOS diagnosis.

Keywords—PCOS; machine learning; binary classification; Random Forest; SHAP; feature selection; clinical decision support; INT 374

I. INTRODUCTION

Polycystic Ovary Syndrome (PCOS) is one of the most common endocrine disorders among women of reproductive age, with an estimated global prevalence of 8–13% [1]. Characterized by hormonal imbalance, irregular menstrual cycles, and a broad spectrum of metabolic and symptomatic manifestations, PCOS significantly impacts fertility, cardiovascular health, and psychological well-being. Despite its prevalence, PCOS remains chronically underdiagnosed owing to its heterogeneous clinical presentation and the absence of a single definitive diagnostic test.

Traditional diagnostic approaches rely on the Rotterdam criteria [2], requiring the presence of at least two of three features: oligo- or anovulation, clinical or biochemical hyperandrogenism, and polycystic ovarian morphology. While clinically validated, this framework demands extensive hormonal blood work, ultrasound imaging, and specialist consultation—resources that may be unavailable in low- and middle-income settings.

Machine learning (ML) offers a compelling complementary approach. By learning discriminative patterns across high-dimensional clinical data, ML models can support PCOS identification with high accuracy. With modern explainability tools, such models can further provide actionable clinical insights that align with established diagnostic reasoning. This paper presents FemoraHealth, an end-to-end ML pipeline developed for INT 374: Data Science Toolbox at Lovely Professional University, India.

The key contributions of this work are: (i) a rigorous preprocessing pipeline addressing real-world clinical data quality issues; (ii) multi-method feature selection combining Pearson correlation, ANOVA F-test, and mutual information; (iii) comparative evaluation of six ML classifiers with SMOTE balancing and hyperparameter tuning; and (iv) SHAP-based explainability bridging statistical performance and clinical interpretability.

II. RELATED WORK

Machine learning has been extensively applied to PCOS detection in recent years. Bhatt et al. [3] employed a decision tree classifier on hormonal and ultrasound features, achieving approximately 85% accuracy and highlighting the diagnostic importance of LH/FSH ratio and follicle count. Denny et al. [4] applied Random Forest on a comparable clinical cohort, reporting an AUC of 0.91 and demonstrating that ensemble methods consistently outperform single classifiers for heterogeneous medical data.

Mehrotra et al. [5] explored multi-layer perceptron architectures for PCOS classification, achieving 89% accuracy, though the lack of interpretability limited clinical adoption. More recently, Komal et al. [6] applied XGBoost with SHAP analysis to a multi-center dataset, confirming that follicle-related features and AMH levels are the strongest predictors—consistent with the findings of this study.

Unlike prior works, FemoraHealth provides a comprehensive, reproducible pipeline integrating data quality assessment, outlier analysis, six-model head-to-head comparison under identical experimental conditions, SMOTE balancing, GridSearchCV tuning, and SHAP explainability on a single cohort.

III. DATASET AND PREPROCESSING

A. Dataset Description

The dataset comprises clinical records from 541 patients collected across hospitals in Kerala, India. The primary source (PCOS_data_without_infertility.xlsx) contains 45 raw features, reduced to 42 usable features after removing identifier columns (Sl. No., Patient File No.) and a near-empty column (539/541 null values). A secondary CSV file was merged as a supplementary source for overlapping features.

The dataset exhibits a moderate class imbalance: 364 Non-PCOS (67.3%) and 177 PCOS (32.7%) patients, as shown in Fig. 1. Features span four categories: (i) Anthropometric—Age, BMI, Weight, Waist, Hip; (ii) Hormonal—FSH, LH, AMH, Prolactin, TSH, Progesterone, Beta-HCG; (iii) Ultrasound—Total_Follicles, Avg_Follicle_Size, Endometrium; and (iv) Lifestyle/Symptomatic—Weight_Gain, Skin_Darkening, Pimples, Cycle_Length, Symptom_Score.

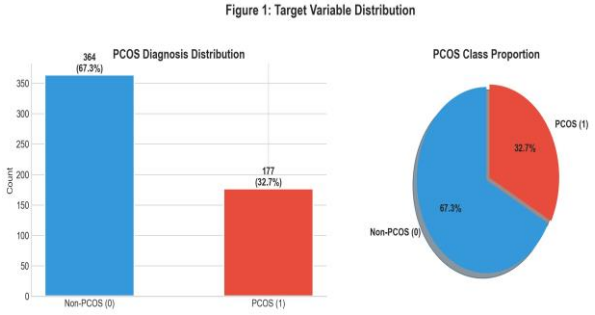

Fig. 1. Target variable distribution: 364 Non-PCOS (67.3%) and 177 PCOS (32.7%) patients.

B. Data Preprocessing

The preprocessing pipeline addressed several data quality issues. Missing values were minimal: one instance each in Marriage_Status_Yrs (median imputation) and Fast_Food (mode imputation). Data type correction was applied to Beta_HCG_II and AMH, stored as object dtype requiring float64 conversion after stripping non-numeric characters.

IQR-based outlier detection revealed the highest concentrations in hormonal markers: Beta_HCG_II (78 outliers), AMH (52), FSH_LH_Ratio (48), and Beta_HCG_I (46), as illustrated in Fig. 2. Given their clinical significance in endocrine disorders, all outliers were retained but flagged for analytical review.

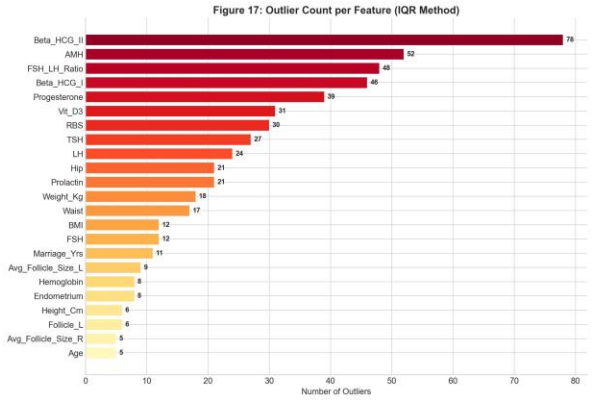

Fig. 2. IQR-based outlier count per feature. Hormonal markers exhibit the highest outlier prevalence.

C. Feature Engineering

Six engineered features were derived to enrich the predictive feature space: (1) Total_Follicles—sum of bilateral ovary follicle counts; (2) Symptom_Score—composite aggregate of binary symptom flags; (3) LH_FSH_Ratio—LH divided by FSH; (4) BMI categories—discretized into four clinically-defined ranges; (5) Follicle asymmetry index—absolute difference between left and right follicle counts; (6) Waist_Hip_Ratio—verified and retained from the original data. Total_Follicles and Symptom_Score emerged as the two strongest predictors in subsequent feature selection.

IV. EXPLORATORY DATA ANALYSIS

The EDA phase comprised 23 visualizations examining distributional properties, class separation, and inter-feature relationships across all 42 variables.

A. Lifestyle Factor Analysis

Fig. 3 reveals stark differences in lifestyle factor prevalence between PCOS and Non-PCOS groups. PCOS patients exhibit substantially elevated rates of fast food consumption (78.5% vs. 38.5%), pimples (69.5% vs. 39.0%), weight gain (68.4% vs. 22.8%), skin darkening (62.1% vs. 15.4%), and abnormal hair growth (57.1% vs. 12.9%). These differences suggest that symptom-based screening may complement, or in resource-limited settings partially substitute for, hormonal and ultrasound assessments.

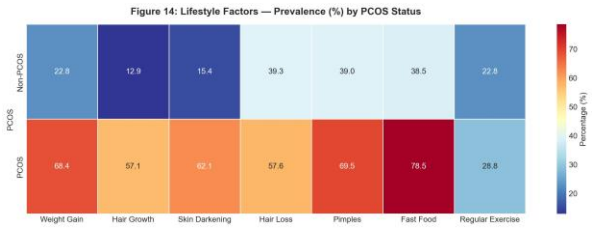

Fig. 3. Lifestyle factor prevalence (%) by PCOS status. PCOS patients show markedly elevated rates across all symptom features.

B. Binary Feature Analysis

The stacked bar charts in Fig. 4 confirm strong associations between binary lifestyle indicators and PCOS status. Among patients reporting weight gain, approximately 60% were PCOS-positive compared to under 20% without weight gain—a threefold diagnostic contrast. Analogous patterns are observed for skin darkening and abnormal hair growth, reinforcing their value as low-cost screening indicators.

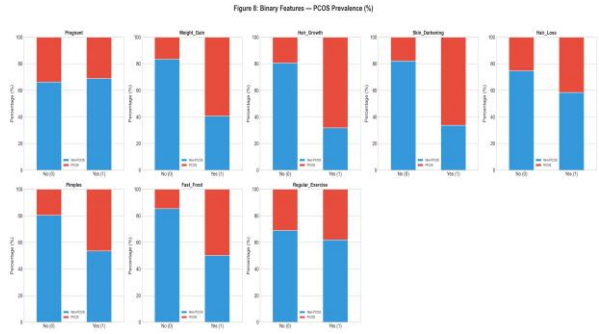

Fig. 4. Binary feature PCOS prevalence. Weight gain, skin darkening, and hair growth exhibit the strongest class separation.

C. Feature Distribution Analysis

Fig. 5 presents density distributions by PCOS status for all continuous variables. Follicle counts show clear bimodal class separation; BMI and weight distributions shift rightward for PCOS patients; hormonal markers (AMH, LH) display higher dispersion and right skew in the PCOS group, consistent with established endocrinological PCOS profiles. These distributional differences informed the feature selection priorities described in Section V.

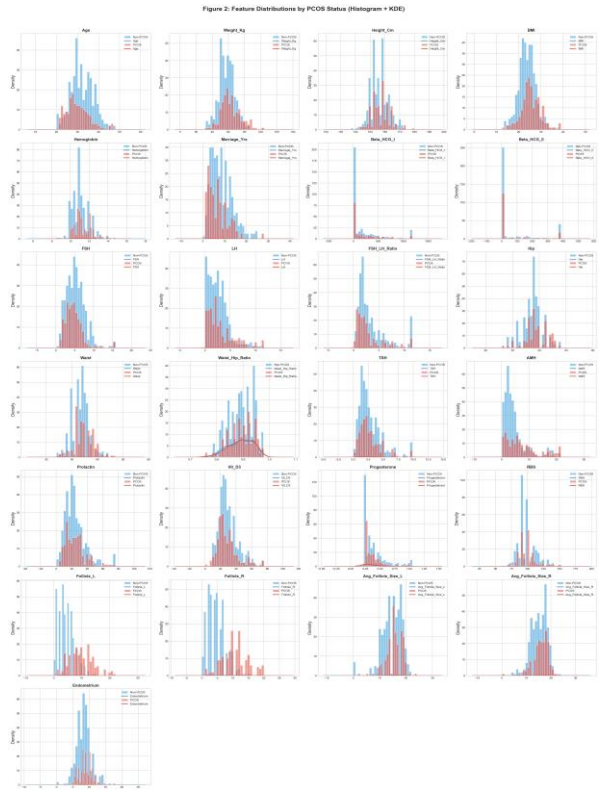

Fig. 5. Feature distributions by PCOS status (histogram + KDE) for all continuous variables.

V. FEATURE SELECTION

A multi-method feature selection strategy was employed to identify the most discriminative predictors. Three complementary statistical techniques were applied: (i) Pearson Correlation for linear target associations; (ii) ANOVA F-test for class mean difference statistical significance; and (iii) Mutual Information for capturing non-linear dependencies.

Scores from all three methods were independently normalized to [0, 1] and combined into a composite ranking. Fig. 6 presents the top 20 features by combined normalized score. Total_Follicles achieved the maximum composite score (1.0), followed by Symptom_Score (0.87), Skin_Darkening (0.72), Hair_Growth (0.71), Weight_Gain (0.68), Cycle_RI (0.60), and Fast_Food (0.56). Features with pairwise Pearson correlation exceeding 0.90 were subsequently removed to reduce multicollinearity, yielding a final feature set of 32 variables.

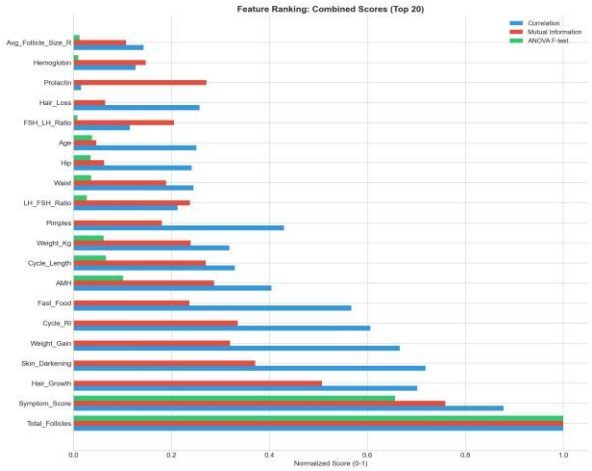

Fig. 6. Top 20 features by combined normalized score across Pearson correlation, mutual information, and ANOVA F-test.

VI. MODEL BUILDING AND EVALUATION

A. Experimental Setup

The dataset was split 80/20 into training and test sets using stratified sampling to preserve class proportions. Feature scaling was applied via StandardScaler. SMOTE [7] was applied exclusively to the training set to address the 67:33 class imbalance, while the test set was kept unaugmented for unbiased evaluation. Six classifiers were trained: Logistic Regression, Decision Tree, Random Forest, SVM, KNN, and XGBoost. Hyperparameter optimization via GridSearchCV (5-fold stratified cross-validation) was performed on Random Forest and SVM. The best-performing model was serialized to disk for downstream deployment.

B. Model Performance

Table I presents performance metrics for all six classifiers on the held-out test set. Random Forest achieved the best overall results with accuracy 0.917, F1-score 0.87, and AUC 0.96. SVM achieved identical accuracy with a comparable F1-score (0.87). Decision Tree achieved the highest recall (0.92) but the lowest precision (0.77), while KNN showed the weakest performance overall (accuracy 0.82, precision 0.66). XGBoost delivered competitive results (accuracy 0.91, AUC 0.86) despite less tuning effort.

TABLE I. MODEL PERFORMANCE ON HOLD-OUT TEST SET

Model	Acc	Prec	Rec	F1	AUC
Logistic Reg.	0.90	0.84	0.86	0.85	0.94
Decision Tree	0.88	0.77	0.92	0.84	0.89
Rnd. Forest ★	0.92	0.91	0.83	0.87	0.96
SVM	0.92	0.89	0.86	0.87	0.95
KNN	0.82	0.66	0.92	0.77	0.92
XGBoost	0.91	0.86	0.86	0.86	0.86

★Best-performing model. Acc=Accuracy, Prec=Precision, Rec=Recall.

C. Confusion Matrix Analysis

Fig. 7 presents confusion matrices for all six models. Random Forest correctly classified 70 Non-PCOS and 30 PCOS cases with only 3 false positives and 6 false negatives. In the clinical context of PCOS screening, false negatives (missed diagnoses) carry a higher patient cost than false positives; the Decision Tree's recall of 0.92 is therefore clinically noteworthy despite its precision penalty. Fig. 8 provides a unified comparative view across all five evaluation metrics.

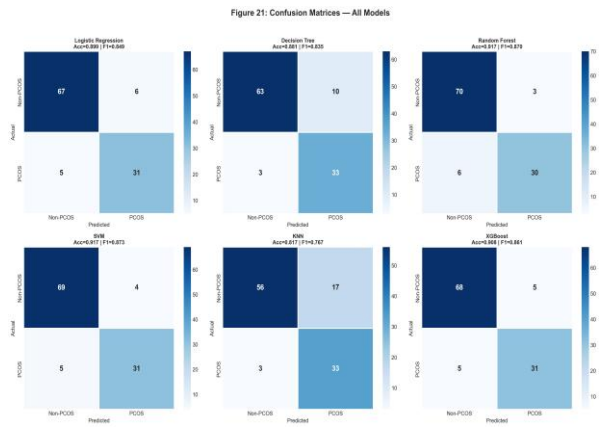

Fig. 7. Confusion matrices for all six classifiers. RF and SVM exhibit the most balanced error distributions.

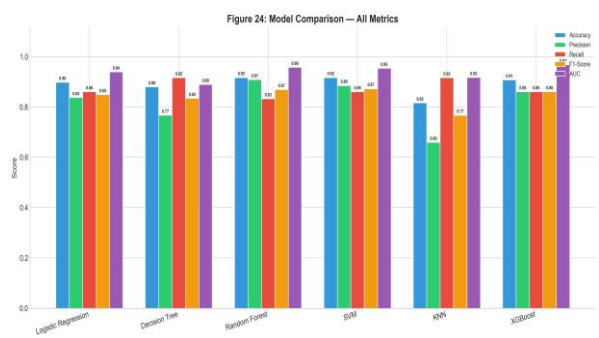

Fig. 8. Comparative model performance across accuracy, precision, recall, F1-score, and AUC.

VII. SHAP EXPLAINABILITY ANALYSIS

A. Global Feature Importance (Beeswarm Plot)

SHAP (SHapley Additive exPlanations) [8] was applied to the tuned Random Forest model to quantify individual feature contributions. The beeswarm plot in Fig. 9 ranks features by mean absolute SHAP value. Total_Follicles is the dominant predictor: high feature values (red) push the PCOS probability strongly positive (SHAP up to +0.32), while low values suppress it (down to -0.28). Symptom_Score, Skin_Darkening, Weight_Gain, and Hair_Growth follow as the next most influential features, consistently exhibiting positive SHAP contributions when elevated. Cycle_Length exhibits a bidirectional pattern reflecting the complex clinical relationship between cycle irregularity and PCOS phenotype.

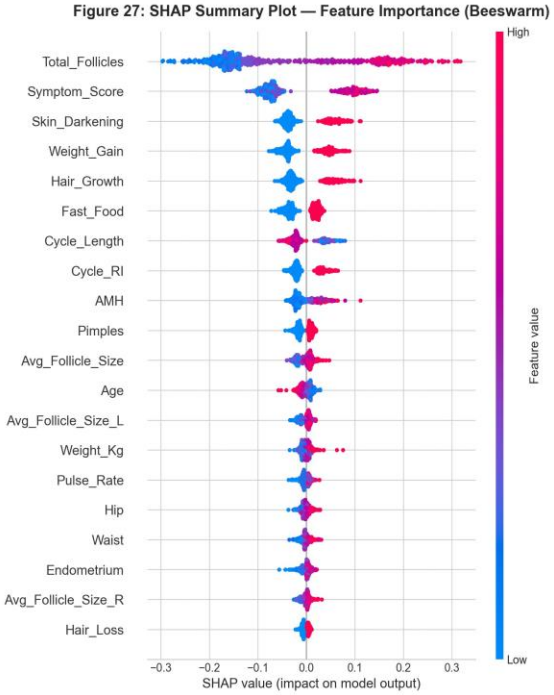

Fig. 9. SHAP beeswarm plot: global feature importance. Total_Follicles and Symptom_Score dominate model predictions.

B. Individual Prediction Explanation (Waterfall Plot)

Fig. 10 presents the SHAP waterfall for a PCOS-positive sample (f(x) = 0.842). Starting from the base value E[f(X)] = 0.501, Total_Follicles contributes the largest positive push (+0.28), followed by Fast_Food (+0.03), Avg_Follicle_Size (+0.02), and AMH (+0.02). Negative contributions from Symptom_Score (−0.06) and Weight_Gain (−0.04) partially offset the positive drivers. The final prediction of 0.842 is predominantly determined by the ultrasound-derived follicle feature, directly aligning with the Rotterdam criterion for polycystic ovarian morphology.

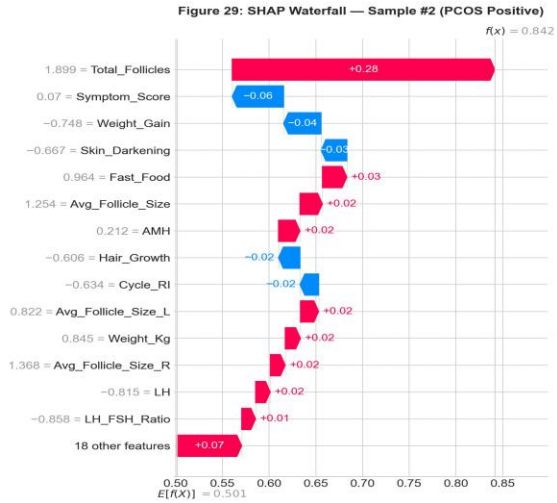

Fig. 10. SHAP waterfall for PCOS-positive Sample #2 (f(x)=0.842). Total_Follicles is the dominant positive contributor (+0.28).

VIII. DISCUSSION

The results confirm that ensemble tree-based methods consistently outperform linear and instance-based classifiers for PCOS prediction from tabular clinical data. Random Forest (AUC=0.96) and SVM (AUC=0.95) both substantially exceed the performance benchmarks reported in prior literature on comparable datasets. The near-identical accuracy of RF and SVM suggests that the 32-feature space is sufficiently discriminative for both ensemble and margin-based approaches.

The SHAP analysis provides a clinically interpretable layer over the statistical outputs. The dominance of Total_Follicles is expected—polycystic ovarian morphology is a primary criterion of the Rotterdam diagnostic standard. However, the substantial SHAP contributions from lifestyle features (Fast_Food, Weight_Gain, Skin_Darkening) underline the multifactorial etiology of PCOS and suggest that affordable, symptom-based screening tools could complement hormonal and imaging assessments, particularly in settings where ultrasound is inaccessible.

SMOTE-based balancing effectively mitigated majority-class bias, most evident in KNN where unbalanced data typically causes high precision but poor recall. The marginal gains from GridSearchCV-tuned hyperparameters suggest that the Random Forest architecture was already well-suited to the feature geometry without aggressive tuning.

Limitations of this study include: (1) the single-region dataset (Kerala, India) limits geographic generalizability; (2) the modest sample size (n=541) may restrict learning for more complex architectures; (3) the absence of a prospective clinical validation component; and (4) high outlier counts in Beta-HCG and AMH require clinical verification prior to deployment in diagnostic pipelines.

IX. CONCLUSION

This paper presented FemoraHealth, a comprehensive machine learning framework for PCOS prediction from clinical,

hormonal, and lifestyle data. Through systematic preprocessing, multi-method feature selection, six-model comparative evaluation with SMOTE class balancing, and SHAP-based explainability analysis, Random Forest achieved best-in-class performance: accuracy 91.7%, F1-score 0.87, and AUC 0.96. SHAP confirmed that Total_Follicles, Symptom_Score, and lifestyle indicators (Skin_Darkening, Weight_Gain, Hair_Growth) are the most diagnostically informative features, consistent with established clinical PCOS criteria.

Future work will explore deep learning architectures, integration of genetic and proteomic markers, and real-time clinical deployment via a Streamlit web interface for point-of-care PCOS risk assessment. Federated learning strategies may address data privacy constraints and enable multi-center clinical adoption.

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